- *Realism of projections*. Croatia's projected fiscal adjustment in the cyclically-adjusted primary balance-to-GDP ratio over the medium term is similar in magnitude to adjustments of other countries.

- *Vulnerabilities*: The heat map shows that the stock of debt remains high. At the same time, the gross financing needs have declined since 2015, and they are no longer above the 15 percent high-risk threshold. The debt profile has also improved in terms of longer maturity, market perception, change in the share of short-term debt, and the share of public debt held by non-residents.

The Baseline Scenario Incorporates

- *Fiscal path*: Preliminary data put the 2019 general government budget balance at around 0 percent of GDP, including the second payment of state guarantees of Uljanik shipyards of about 0.3 percent of GDP.

- *Growth*: Real GDP growth is estimated at 2.9 percent for 2019. Going forward, growth is expected to converge towards its potential rate of about 2.0 percent.

Risks to the baseline come mainly from potentially less favorable growth prospects but past track record of fiscal overperformance mitigates the balance of risks.

- While Croatia's real GDP growth forecast has been optimistic since 2009, errors have diminished in magnitude in recent years, and growth has even surprised on the upside. The growth shock could come from external risks identified in the RAM (Annex V), reflecting reduced demand from trading partners and a slowdown of tourism activity.

- A neutral interest rate-growth differential and the consolidation efforts could result in much faster debt contraction than previously projected. Gross financing needs are also projected to decline due to lower deficits and recent refinancing of debt at longer maturities and favorable interest rates, resulting in lower effective rates even after accounting for a gradual normalization of global monetary conditions.

Shocks and Stress Tests

- **Slower output growth is the main risk to debt sustainability.** Underperformance of real GDP growth by one standard deviation results in the primary deficit deteriorating in 2022 to about -0.9 percent of GDP and debt sharply increasing to over 76.7 percent, before improving thereafter. Gross financing needs would be around 13.8 percent of GDP in 2021 and decline slowly going forward.

A combination of macro-fiscal shocks further highlights the sensitivity of public debt and gross financing needs to adverse scenarios. If shocks to real GDP growth, the primary balance, real exchange rate, and real interest rate were to occur simultaneously, public debt would increase sharply and reach nearly 77 percent of GDP (or about 165 percent of public revenues) at its peak, and gross financing needs would be close to 14 percent of GDP in 2021–22.

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Figure 1. Croatia: Public Sector Sustainability Analysis (DSA) – Baseline Scenario

(In percent of GDP, unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

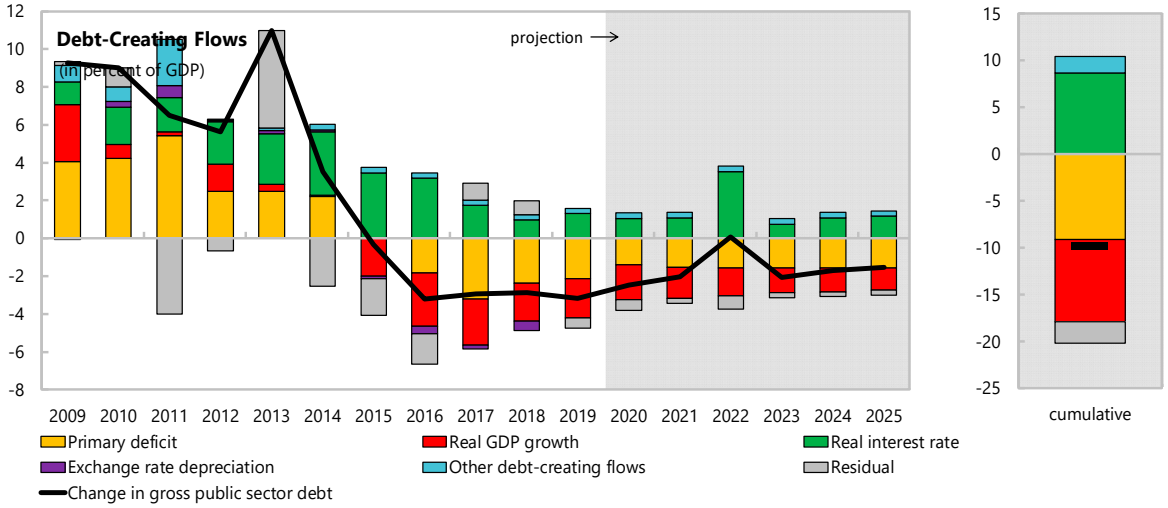
	Actual			Projections					
	2009-2017 [2/]	2018	2019	2020	2021	2022	2023	2024	2025
Nominal gross public debt	71.7	74.7	71.5	69.1	67.0	67.1	65.0	63.3	61.8
Public gross financing needs	15.3	6.5	8.9	11.4	11.5	8.4	8.5	8.1	9.8
Real GDP growth (in percent)	-0.3	2.7	2.9	2.7	2.5	2.2	2.1	2.0	2.0
Inflation (GDP deflator, in percent)	1.0	1.8	1.4	1.6	1.7	1.7	2.0	2.1	2.0
Nominal GDP growth (in percent)	0.6	4.5	4.4	4.3	4.3	3.9	4.1	4.2	4.0
Effective interest rate (in percent) [4/]	4.5	3.1	3.3	3.2	3.4	3.4	3.5	3.7	3.8

As of January 24, 2020

Sovereign Spreads		
EMBIG (bp) 3/		93
5Y CDS (bp)		46
Ratings	Foreign	Local
Moody's	Ba2	Ba2
S&Ps	BBB-	BBB-
Fitch	BBB-	BBB-

Contribution to Changes in Public Debt

	Actual			Projections							
	2009-2017	2018	2019	2020	2021	2022	2023	2024	2025	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	4.3	-2.9	-3.2	-2.5	-2.1	0.1	-2.1	-1.7	-1.5	-9.8	
Identified debt-creating flows	4.7	-3.6	-2.6	-1.9	-1.8	0.8	-1.8	-1.5	-1.3	-7.5	
Primary deficit	1.8	-2.4	-2.1	-1.4	-1.5	-1.6	-1.6	-1.6	-1.6	-9.1	0.3
Automatic debt dynamics [5/]	2.3	-1.5	-0.8	-0.8	-0.6	2.0	-0.6	-0.2	0.0	-0.1	
Interest rate/growth differential [6/]	2.2	-1.0	-0.8	-0.8	-0.6	2.0	-0.6	-0.2	0.0	-0.1	
Exchange rate depreciation [7/]	0.1	-0.5	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.6	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	1.7	
Residual, including asset changes [8/]	-0.4	0.7	-0.5	-0.6	-0.2	-0.7	-0.2	-0.2	-0.2	-2.3	



Source: IMF staff.

1/ Public sector is defined as general government.

2/ Based on available data.

3/ Long-term bond spread over German bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

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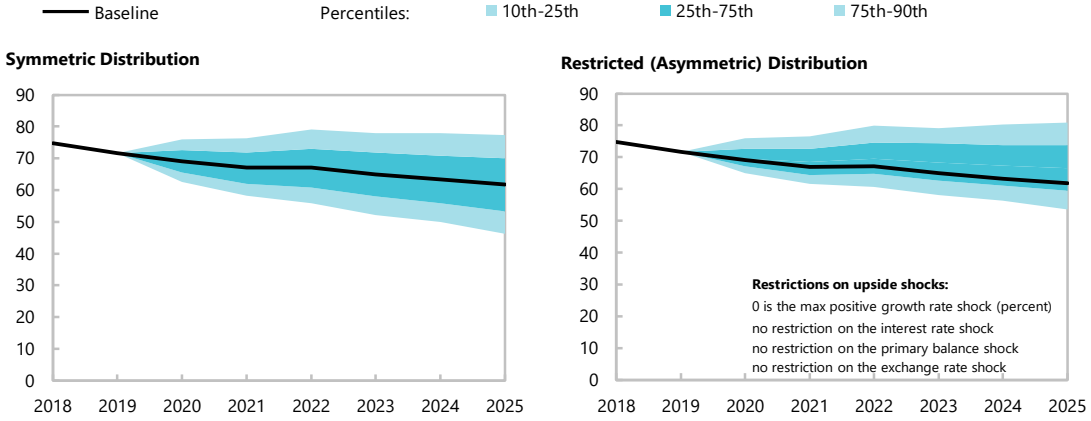
Figure 2. Croatia: Public DSA Risk Assessment

Heat Map

Debt level [1/]	Real GDP Growth Shock	Primary Balance Shock	Real Interest Rate Shock	Exchange Rate Shock	Contingent Liability shock
Gross financing needs [2/]	Real GDP Growth Shock	Primary Balance Shock	Real Interest Rate Shock	Exchange Rate Shock	Contingent Liability Shock
Debt profile [3/]	Market Perception	External Financing Requirements	Change in the Share of Short-Term Debt	Public Debt Held by Non-Residents	Foreign Currency Debt

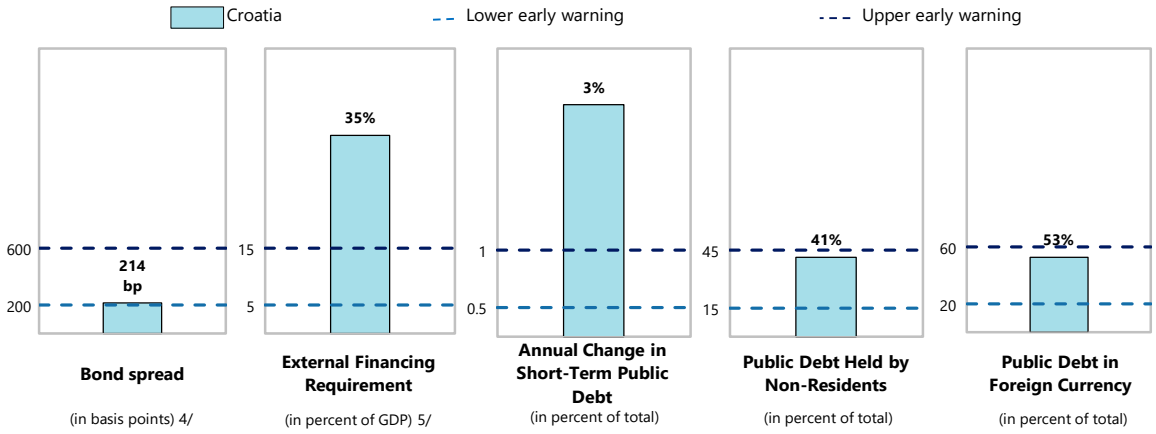
Evolution of Predictive Densities of Gross Nominal Public Debt

(in percent of GDP)



Debt Profile Vulnerabilities

(Indicators vis-à-vis risk assessment benchmarks, in 2019)



Source: IMF staff.

1/ The cell is highlighted in green if debt burden benchmark of 70% is not exceeded under the specific shock or baseline, yellow if exceeded under specific shock but not baseline, red if benchmark is exceeded under baseline, white if stress test is not relevant.

2/ The cell is highlighted in green if gross financing needs benchmark of 15% is not exceeded under the specific shock or baseline, yellow if exceeded under specific shock but not baseline, red if benchmark is exceeded under baseline, white if stress test is not relevant.

3/ The cell is highlighted in green if country value is less than the lower risk-assessment benchmark, red if country value exceeds the upper risk-assessment benchmark, yellow if country value is between the lower and upper risk-assessment benchmarks. If data are unavailable or indicator is not relevant, cell is white. Lower and upper risk-200 and 600 basis points for bond spreads; 5 and 15 percent of GDP for external financing requirement; 0.5 and 1 percent for change in the share of short-term debt; 15 and 45 percent for the public debt held by non-residents; and 20 and 60 percent for the share of foreign-currency denominated debt.

4/ Long-term bond spread over German bonds, an average over the last 3 months, 26-Oct-19 through 24-Jan-20.

5/ External financing requirement is defined as the sum of current account deficit, amortization of medium and long-term total external debt, and short-term total external debt at the end of previous period.

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